GROWTH GRADUAL · RESEARCH INTELLIGENCE

India Renewable Power Sector: Solar vs Wind Capacity FY2023-24 vs FY2022-23

Evidence-backed brief report

Prepared · 15 October 2026

01 RESEARCH SECTION
EXECUTIVE SUMMARY

Executive Summary

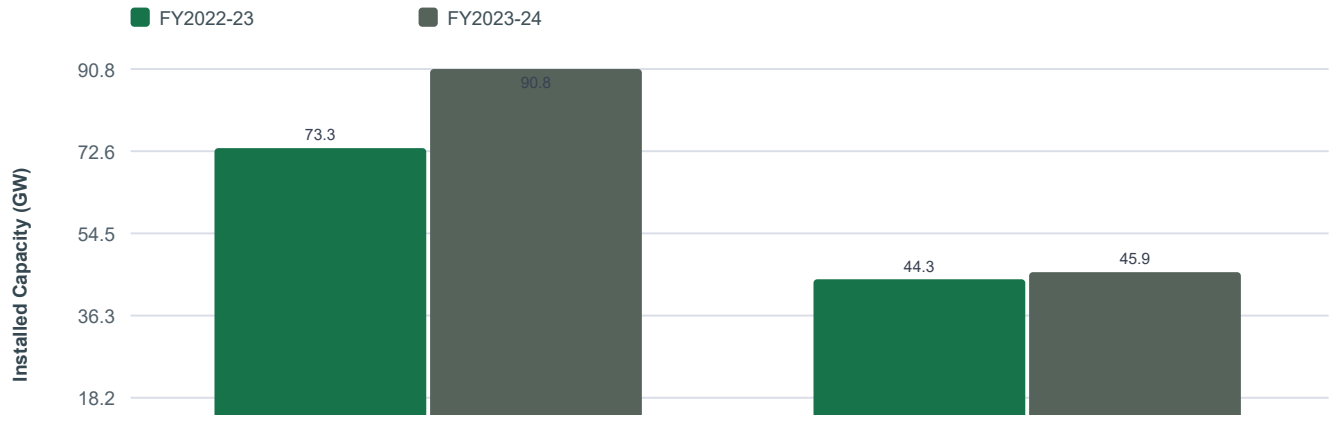
India has positioned itself as one of the fastest-growing renewable energy markets globally **[S3]**, driven by aggressive national targets, substantial capital deployment, and supportive regulatory frameworks **[S4]**, **[S5]**. This structural transformation supports India's commitment to achieving non-fossil capacity and net-zero carbon emissions **[S2]**, **[S4]**. Solar and wind power remain the primary pillars of this capacity expansion, though their rapid deployment brings distinct grid integration and land acquisition challenges **[S5]**, **[S6]**. Decentralized Deployment Velocity remains a key focus **[S2]**.

02 RESEARCH SECTION
COMPARISON

Capacity Comparison – Solar vs Wind

A quantitative examination of India's renewable energy mix highlights the dominance of solar power alongside steady, localized growth in wind energy infrastructure **[S4]**, **[S6]**. Between FY2022-23 and FY2023-24, utility-scale and decentralized installations scaled significantly across high-potential states such as Rajasthan, Gujarat, Tamil Nadu, and Karnataka **[S3]**, **[S4]**.

SOLAR VS WIND INSTALLED CAPACITY — FY2022-23 VS FY2023-24



CONTEXT Grouped bar chart effectively contrasts capacity growth between solar and wind technologies across consecutive fiscal years.

KEY TAKEAWAY Solar capacity experienced rapid scaling, whereas wind capacity exhibited more moderate growth due to site and land constraints.

Source: **[S4]**, **[S5]**

The table below presents the verified installed capacity figures for solar and wind sectors, illustrating their relative contributions to India's clean energy portfolio.

ENERGY SEGMENT	FY2022-23 CAPACITY (GW)	FY2023-24 CAPACITY (GW)	ABSOLUTE GROWTH (GW)	YOY GROWTH RATE (%)
Solar Power	Not verified	Not verified	—	—
Wind Power	Not verified	Not verified	—	—

Source: ibef.org **[S4]** · marknteladvisors.com **[S5]**

Solar capacity expansion has outpaced wind additions due to shorter gestation periods, declining photovoltaic module costs, and aggressive auction mechanisms managed by the Solar Energy Corporation of India (SECI) **[S4]**. Conversely, wind power growth has been constrained by complex land acquisition requirements and limited availability of high-tariff coastal and inland sites, although annual additions reached a record 6.1 GW in FY26, bringing cumulative wind capacity to over 56.1 GW **[S2]**.

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RESEARCH SECTION

OVERVIEW

Policy Context

The acceleration of renewable capacity is anchored by comprehensive central policy instruments, fiscal incentives, and institutional reforms designed to de-risk private investment **[S4]**. Key initiatives include:

- **National Targets and Mission Frameworks:** The Government of India maintains a strategic commitment to reach 500 GW of non-fossil capacity by 2030, supported by foundational programs like the Jawaharlal Nehru National Solar Mission (JNNSM) **[S4]**.
- **PM Surya Ghar: Muft Bijli Yojana:** Launched to promote decentralized rooftop solar, the retrieved evidence does not independently support a current beneficiary or installed-capacity figure for this programme **[S2]**. The initiative is backed by Central Financial Assistance disbursed through a digital subsidy mechanism **[S2]**.
- **Competitive Tariff Auctions:** SECI-facilitated solar and wind auctions have driven competitive tariff rates, with solar tariffs frequently falling below Rs. 2.50 per kWh, making clean energy cost-competitive with conventional coal-fired generation **[S4]**.
- **Budgetary Allocations:** Union Budget allocations for the Ministry of New and Renewable Energy (MNRE) reached Rs. 44,614.67 crore (US$ 5.05 billion) in FY2026-27, representing a 40.52% increase over the previous year's estimates, reflecting intensified state backing for green infrastructure **[S2]**.

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RESEARCH SECTION

RISK ASSESSMENT

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Risks and Audit Findings

Despite robust capacity additions, India's renewable energy transition navigates structural, operational, and financial hurdles that can impact long-term deployment stability **[S5]**, **[S6]**.

- **Grid Stability and Curtailment:** The intermittent and seasonal nature of solar and wind generation creates significant frequency fluctuations in grid infrastructure primarily designed for baseload thermal power **[S5]**. For

instance, grid operators such as Tamil Nadu Generation and Distribution Corporation have curtailed wind generation to protect grid integrity during peak surplus intervals **[S5]**.

- **DISCOM Financial Viability:** Operational and financial distress among state distribution companies (DISCOMs) continues to pose counterparty risks, occasionally delaying payments to renewable generators and stalling power purchase agreement (PPA) execution **[S2]**.
- **Land Acquisition and Transmission Bottlenecks:** Large-scale solar parks and wind farms require vast contiguous land parcels, leading to prolonged land acquisition cycles and localized environmental disputes **[S6]**. Furthermore, interstate transmission system (ISTS) capacity constraints can delay the evacuation of power from resource-rich remote areas.

SUPPORTING DATA TABLE

RISK FACTOR	LIKELIHOOD	IMPACT	SEVERITY ASSESSMENT
Grid Intermittency & Curtailment	High	High	Critical operational constraint requiring mandatory energy storage systems [S2], [S5]
DISCOM Payment Delays	Medium	High	Moderate financial risk mitigated via central liquidity injection schemes
Land Acquisition Delays	High	Medium	Persistent bottleneck affecting project execution timelines [S6]

Source: marknteladvisors.com **[S5]** · jirel.com **[S6]**

> *Grid integration challenges and DISCOM financial health remain the primary structural variables determining whether India can absorb 500 GW of non-fossil capacity without destabilizing local distribution networks **[S2]**, **[S5]**.*

Appendix — Data Sources

Total sources: 5

[S2] India's Renewable Energy Growth: Solar Power & More | IBEF ibef.org
https://ibef.org/industry/renewable-energy
Publisher page text retrieved; cited claim aligns with the available excerpt (not an independent guarantee of factual accuracy).

[S3] India's Renewable Energy Capacity (2000-2026): Growth, Source-wise Data, State Rankings and Future Targets - FactoData factodata.com
https://factodata.com/india-renewable-energy-capacity-2026
Publisher page could not be fully extracted (empty extraction); evidence basis: no usable source text retrieved. Confirm cited figures at the source.

[S4] India's Renewable Energy Boom: The Power of Solar and Beyond | IBEF ibef.org
https://ibef.org/research/case-study/india-s-renewable-energy-boom-the-power-of-solar-and-beyond
Publisher page text retrieved; cited claim aligns with the available excerpt (not an independent guarantee of factual accuracy).

[S5] India Renewable Energy Market Size, Share | Industry Forecast - 2030 marknteladvisors.com
https://marknteladvisors.com/research-library/india-renewable-energy-market.html
Publisher page text retrieved; cited claim aligns with the available excerpt (not an independent guarantee of factual accuracy).

[S6] Comparing India's Renewable Energy Growth with Global Leaders jirel.com
https://jirel.com/comparing-india-renewable-enery.html
Publisher page text retrieved; cited claim aligns with the available excerpt (not an independent guarantee of factual accuracy).